ANANYA ROY

MSc BIOTECHNOLOGY | FOOD BIOTECHNOLOGY

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MSc Biotechnology student at Wageningen University with hands-on experience in microfluidic device development, molecular biology, tissue culture and vaccine quality control. Interested in cultivated meat, cellular agriculture and precision fermentation.

EDUCATION

Wageningen University and Research (WUR), Netherlands *Sep 2025 - present*

MSc Biotechnology (Specialization Food) | Grade 7.55

Vellore Institute of Technology (VIT), Vellore *Sep 2021 – Aug 2025*

BTech Biotechnology | CGPA 9.14

RESEARCH & EXPERIENCE

Master Thesis at Wageningen University and Research (WUR) *Sep 2026 - present*

Cell line optimization of white body's cells from Octopus vulgaris.

Researching conditions for the establishment of both short term and long-term cell culture of the octopus' white body cells in vitro. Working on establishing pipelines for cell viability using Fluorescence Microscopy, Flow Cytometry and ImageJ. Will also be testing for proliferation using the EdU kit and cell characterization using different stains and PCR for species identification.

Bachelor Thesis at Indian Institute of Technology (IIT), Madras *5th Jan 2025 – 5th May 2025*

Microfluidic Device for Exosome Filtration from chemically stimulated chicken skeletal muscles

Focused on the development and optimization of a microfluidic device using polydimethylsiloxane (PDMS) with microscale channels to separate exosomes from chemically stimulated chicken muscles. Gained hands on training on dissection processes involving chicken muscles. Worked with autoCAD to design the channels, used corona treatment to bond PDMS to glass layers. Used DiI Staining to analyze the muscle fibers and S-eye software and ImageJ to visualize the muscle fibers.

Summer Intern at Heartfulness Tree Conservation Center, Hyderabad *10th Jun 2024 – 29th June 2024*

Plant Tissue Culture and Micropropagation for plant cultivation and propagation

Completed an internship focused on advanced micropropagation techniques and plant tissue culture of various species. Gained in-depth experience in the cultivation and propagation of plants through tissue culture methods.

Intern at Serum Institute of India, Pune *7th Aug 2023 – 15th Dec 2023*

Improving effectiveness of pneumococcal vaccines using Biotinylation

Worked on developing a multiplexed assay that could detect antibodies against multiple pneumococcal serotypes in a single experiment compared to conventional methods that test each serotype individually. Optimized the attachment of biotin to the pneumococcal polysaccharides so that they could bind to streptavidin- coated assay plates. Used Meso-Scale Discovery (MSD) Technology that detects antibody binding via electrochemiluminescence.

TECHNICAL SKILLS

Animal Cell Culture | Plant Tissue Culture | PDMS handling and fabrication | AutoCAD | Corona Treatment | Polysaccharide Biotinylation | PCR | Meso Scale Discovery (MSD) Technology | Fluorescence Microscopy | Confocal | ImageJ | GraphPad Prism | Flow Cytometry | Java | Python | Excel

PUBLICATION

Published a first author review article in Taylor & Francis: "Emerging Contaminants Jeopardizing Groundwater Integrity: Environmental Implications."

DOI: 10.1080/26395940.2025.2572634

AWARDS & LEADERSHIP

Biobased Innovation Student Challenge Europe (BISC-E)

Innovation surrounding willow pruning waste to turn them into bioactive compounds and biodegradable mycelium materials

Best Business Model Award (2023) – IGNITIA Hackathon, VIT

Innovation surrounding technology for alleviating menstrual cramps

Second Prize – BIDFest (Bio-Inspired Design Fest), VIT

Environmental and biomedical waste treatment concept

Vice-Chairperson, Culinary Club (Feb 2023 – Jan 2024)

Coordinated club activities and events and served as an event representative for 2 university events.

ADDITIONAL EXPERIENCE & LANGUAGES

Part-time waitress at Veg Nation, Utrecht *20th March 2026 - present*

Languages – English (C2), Dutch (A1)
